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Development of an Integrated System for Acoustic Calibration of Stethoscopes

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Abstract

Stethoscopes may undergo material degradation that alters acoustic transmission and attenuates clinically relevant cardiopulmonary sounds. This project aimed to develop an integrated acoustic calibration system for characterizing stethoscope frequency response and signal fidelity, generating controlled signals across the 20–2000 Hz cardiopulmonary bandwidth, acquiring high-fidelity audio, and supporting repeatable spectral analysis.

Controlled sinusoidal tests demonstrated capture of 20 Hz and 50 Hz signals, with concentrated spectral energy and generally stable amplitude envelopes, although amplitude surges and noise-floor changes were observed. At 200 Hz, the spectrogram showed harmonic components at integer multiples of the fundamental frequency, indicating total harmonic distortion. An 1111 Hz test produced a continuous spectral band without attenuation or digital blindness. The optimized acquisition rate remained well above the 4000 Hz Nyquist threshold required for the 2000 Hz bandwidth.

The developed system demonstrated functional low- and high-frequency acquisition across the intended cardiopulmonary range. However, observed gain-related amplitude variations and harmonic distortion indicate that further refinement is required before complete stethoscope calibration performance can be established.

Keywords: Calibração de estetoscópios, transdutor acústico, processamento digital de sinais

Resumo

Os estetoscópios podem sofrer degradação do material que altera a transmissão acústica e atenua os sons cardiopulmonares clinicamente relevantes. Este projeto teve como objetivo desenvolver um sistema integrado de calibração acústica para caracterizar a resposta em frequência e a fidelidade do sinal do estetoscópio, gerando sinais controlados na largura de banda cardiopulmonar de 20–2000 Hz, adquirindo áudio de alta fidelidade e permitindo uma análise espectral repetível.

Testes sinusoidais controlados demonstraram a captura de sinais de 20 Hz e 50 Hz, com energia espectral concentrada e envelopes de amplitude geralmente estáveis, embora tenham sido observados picos de amplitude e variações no ruído de fundo. A 200 Hz, o espectrograma revelou componentes harmónicos em múltiplos inteiros da frequência fundamental, indicando distorção harmónica total. Um teste a 1111 Hz produziu uma banda espectral contínua sem atenuação nem cegueira digital. A taxa de aquisição otimizada manteve-se bem acima do limiar de Nyquist de 4000 Hz exigido para a largura de banda de 2000 Hz.

O sistema desenvolvido demonstrou aquisição funcional de baixas e altas frequências em toda a gama cardiopulmonar pretendida. No entanto, as variações de amplitude relacionadas com o ganho e a distorção harmónica observadas indicam que é necessário um maior aperfeiçoamento antes de se poder estabelecer um desempenho completo de calibração do estetoscópio.

Palavras-Chave: Stethoscope Calibration, Acoustic Transducer, Digital Signal Processing

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List of Symbols

Symbol	Description	Units
f	frequência	Hz
f	força	N
f_s	frequência de amostragem	Hz
i	corrente	A
M	massa	kg
P	potência	W
t	tempo	s
V	tensão elétrica	V
x	deslocamento	m
θ	deslocamento angular	rad
ω	velocidade angular	$\mathrm{rad\,s^{-1}}$

Chapter 1

Introduction

1.1 State of the Art/Contextualization

Stethoscopes remain the most fundamental and prevasive diagnostic tools in clinical practice for the auscultation of cardiopulmonary and vascular sounds. Currently, traditional acoustic stethoscopes dominate over 90% of the global market due to their high durability, low cost, and independence from electrical power sources [1]. However, purely acoustic auscultation is inherently limited not only by the fallibility of human auditory perception but also by the unrecognized physical degradation of the instrument over time. Studies have demonstrated that general practitioners miss up to 57% of significant valvular heart diseases when relying on standard stethoscopes [2].

While frequently attributed to human auditory limitations, this high error rate may be significantly composed by unnoticed material failures; such as compromised acoustic seals in the eartips, hardened tubing, or worn diaphragms. These physical degradations silently alter the stethoscope's frequency response, attenuating vital low-frequency biological sounds before they ever reach the clinician's ear. To mitigate these human diagnostic errors, the medical field is experiencing the gradual emergence of electronic stethoscopes designed to amplify, filter, and digitize biological sounds.

The clinical consequences of an uncalibrated or degraded stethoscope are severe. When physical degradation, such as hardened acoustic tubing or compromised eartip seals, artificially attenuates low-frequency biological signals, critical acoustic

biomarkers like the S3 gallop (a primary indicator of heart failure for older adults) may fall below the threshold of hearing. This mech nical attenuation directly leads to false-negative diagnoses, delayed medical interventions, and compromised patient safety. Therefore, establishing a reliable calibration system is not merely a matter of instrument maintenance, but a critical requirement for ensuring diagnostic integrity.

1.2 Project Description

This project proposes the development of an acoustic calibration system designed to characterize the frequency response, and signal fidelity of stethoscopes. The proposed architecture consists of two primary domains: a physical hardware module for sound generation and detection, and a software platform for control, data acquisition, and spectral analysis.

1.2.1 Objectives

By the end of the project, the developed system should be able to:

- **Simulate Cardiopulmonary Acoustics:** Generate controlled acoustic signals that mimic the frequency spectrum of biological sounds, particularly in the frequency range for cardiopulmonary auscultation.
- **Capture High-Fidelity Audio Data:** Accurately measure the stethoscope's output signal using a high-fidelity microphone and data acquisition system, ensuring that the measurement setup itself does not introduce significant distortions.
- **Analyze Frequency Response:** Analyze the acquired audio data to compute the stethoscope's frequency response curve utilizing python-based spectral analysis techniques.
- **Establish a Repeatable Protocol:** Develop a standardized testing protocol that can be easily replicated for routine stethoscope calibration in clinical settings, ensuring consistent and reliable performance assessments.

Chapter 2

Literature Review

2.1 Mechanics of Material Breakdown in Clinical Environments

The polyvinyl chloride (PVC) tubing standard in acoustic stethoscopes is highly susceptible to chemical degradation, specifically hardening and micro-cracking, which is accelerated by frequent exposure to human skin lipids, hair follicle oils, and clinical sanitizing agents .

To maintain strict infection control, hospitals require the continuous disinfection of medical equipment between patient encounters. Clinicians typically utilize alcohol wipes or chlorine-based disinfectants to eliminate surface pathogens. While these chemical agents are effective at neutralizing bacteria and viruses, they are fundamentally detrimental to the structural integrity of soft plastics. Isopropyl alcohol acts as a chemical solvent, and when applied repeatedly to stethoscope tubing, it actively strips the plasticizers out of the PVC polymer matrix. Plasticizers are the specific chemical additives responsible for giving PVC its flexibility and rubber-like elasticity. As these compounds leach out of the material over months of daily cleaning, the tubing inevitably becomes stiff and rigid.

This chemical breakdown is further compounded by the standard physical habits of medical professionals. Clinicians frequently carry their stethoscopes draped around their bare necks during long shifts. Human skin naturally secretes sebum. When the PVC tubing rests directly against the skin, it absorbs these lipid compounds. The

interaction between human skin oils and the plastic surface catalyzes a secondary form of chemical degradation, accelerating the hardening process at the exact physical apex of the tube.

The combination of harsh solvent evaporation from daily sanitation protocols and continuous lipid absorption creates a compounding cycle of material fatigue. As the PVC loses its elasticity and hardens, it loses its ability to flex without sustaining structural damage. Over time, the constant bending of the rigid tubing causes microscopic fractures to form along the surface, directly paving the way for acoustic failure [3].

2.2 Acoustic Principles and Diagnostic Risks of Stethoscope Degradation

As the mechanical compliance of the PVC decreases, the acoustic pathway is fundamentally altered, actively dampening the transmission of subtle diagnostic markers such as S3/S4 gallops or high-frequency pulmonary crackles.

Analog stethoscopes operate on a strict principle of acoustic pressure transmission. The device captures sound waves at the chest piece and channels them directly to the listener's earpieces through a completely sealed column of air [4]. The physical integrity of the PVC tubing is the most critical variable in this transmission process. When the tubing is factory-new and highly flexible, its thick walls serve a dual purpose: they seamlessly reflect the internal pressure waves up the tube while simultaneously acting as a dampening barrier that absorbs and rejects external ambient room noise.

When the PVC loses its plasticizers and hardens, the acoustic physics of the tube change drastically. A rigid polymer can no longer absorb external vibrations effectively. In a bustling clinical setting, ambient noise from alarms, medical equipment, and background conversations easily penetrates the hardened tubing walls. This external acoustic interference mixes with the internal air column, drastically lowering the signal-to-noise ratio and masking the faint frequencies of the patient's biological signals [4].

The most severe diagnostic risk occurs when the hardened tubing eventually develops micro-cracks. These microscopic fissures physically break the hermetic seal of the internal air column. As the cardiopulmonary pressure waves travel upward from the chest piece, acoustic energy leaks out through these fractures into the surrounding environment. This results in severe signal attenuation, meaning the overall volume of the biological sound is drastically reduced by the time it reaches the clinician. This physical loss of acoustic pressure explains precisely why the chemical

breakdown of PVC translates directly into diagnostic failure, as the degraded material science directly compromises the physician's ability to hear clinically significant, low-intensity physiological sounds [5].

Chapter 3

Methodology

3.1 System Requirements

To design an integrated calibration rig capable of characterizing acoustic and electronic stethoscopes, the system must satisfy a rigorous set of functional requirements (defining operational capabilities) and non-functional requirements (defining performance constraints and signal fidelity).

3.1.1 Functional Requirements

The functional requirements outline the necessary signal generation, mechanical coupling, and digital processing capabilities of the calibration rig, as detailed in Table 3.1.

Table 3.1: Functional requirements for the integrated system.

Requirement	**Engineering Description**
Controlled Acoustic Excitation	The system must generate continuous logarithmic sine sweeps covering the full cardiopulmonary frequency spectrum (20 Hz to 2000 Hz).
Structure-Borne Vibration	Acoustic energy must be delivered via a surface exciter mechanically coupled to an elastomeric tissue-simulating medium, avoiding free-air acoustic impedance mismatch.
Acoustic Sealing & Acquisition	The acquisition sensor must form an airtight seal with the stethoscope eartip to prevent low-frequency attenuation and accurately capture cardiopulmonary acoustics down to 20 Hz.
Signal Pre-Amplification	The low-amplitude analog output from the microphone capsule must be buffered and amplified to line-level voltages without clipping or harmonic distortion.
Digital Signal Processing (DSP)	The Central Control PC must acquire the digitized audio stream, compute the Fast Fourier Transform (FFT), and derive the system's frequency response curve.

3.1.2 Non-Functional Requirements

Non-functional requirements dictate the electronic precision, noise limits, and sampling criteria necessary to ensure clinical relevance and measurement repeatability, as presented in Table 3.2.

Table 3.2: Non-functional performance constraints and hardware specifications.

Requirement	**Engineering Description**
Low Total Harmonic Distortion	The power amplification stage must maintain high linearity across the target spectrum without introducing high-frequency switching noise, justifying a Class-AB analog topology.
Ultra-Low Input Noise Floor	The pre-amplifier stage must introduce minimal equivalent input noise ($e_n \leq 5$ nV/$\sqrt{\text{Hz}}$) to resolve faint biological sounds such as pulmonary crackles.
Nyquist-Compliant Sampling	The ADC stage must sample at a minimum rate of 44.1 kHz with at least 12-bit resolution to prevent quantization error and aliasing.
Measurement Repeatability	The physical coupler and elastomer medium must allow different stethoscope models to be tested under identical mechanical loading conditions.

3.2 System Architecture

To satisfy the functional and non-functional requirements established in Section 3.1, the calibration rig is divided into three interconnected domains: the *Sound Generation Stage*, the *Mechanical Domain*, and the *Sound Acquisition Stage*. Figure 3.1 illustrates the complete electromechanical signal path of the proposed architecture.

The signal path originates at the **Central Control PC**, where a digital reference waveform is converted to an analog signal (V_{out}) via a digital-to-analog converter (DAC). This signal drives a custom discrete Class-AB power amplifier based on the TDA2030 integrated circuit, which provides the linear current necessary to excite the Dayton Audio DAEX25 surface exciter without introducing high-frequency switching artifacts.

Within the *Mechanical Domain*, the exciter transfers structure-borne vibrations through a silicone elastomer phantom into the stethoscope's chestpiece. The acoustic wave travels through the stethoscope's internal acoustic tube to the eartip, where it is captured by a -40 dB Electret Condenser Microphone (ECM) capsule housed within a compliant physical coupler designed to ensure an airtight acoustic seal.

Finally, in the *Sound Acquisition Stage*, the microvolt-level signal produced by the ECM capsule is conditioned by an ultra-low-noise pre-amplifier circuit. The boosted line-level signal is digitized by a dedicated 12-bit USB Analog-to-Digital Converter and transmitted back to the Central Control PC for Fast Fourier Transform (FFT) and spectral analysis.

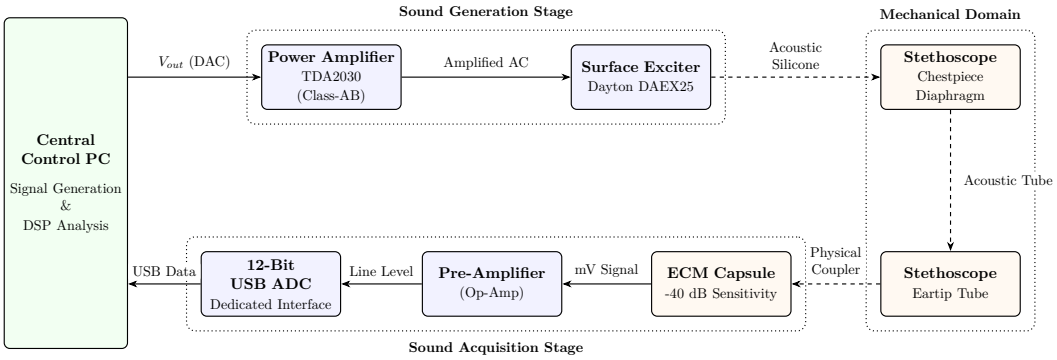


Figure 3.1: Block diagram of the proposed physical rig

3.3 Component Selection

3.3.1 Sound Generation Technologies

Traditional audio testing frequently relies on standard dynamic speakers. However, dynamic speakers operate by displacing free air.

Because a medical stethoscope is clinically designed to capture vibrations through high-impedance solid mediums (such as human tissue and fluid), utilizing an air-coupled speaker introduces significant energy loss, particularly in the low-frequency cardiovascular bands (20 Hz - 150 Hz).

To overcome this limitation, modern biomedical testing methodologies require the use of surface transducers or exciters. Instead of pushing air, surface transducers are designed to directly vibrate the solid object to which they are attached. When a surface transducer is mechanically coupled with a tissue-simulating medium (such as silicone or acoustic gelatine), it can accurately replicate the acoustic impedance of a human chest [6].

Following the justification for structure-borne sound converters, three commercially available surface transducers were selected for evaluation in this project:

- **Dayton Audio TT25-8 PUCK:** This is a compact, high-performance tactile transducer designed for low-frequency applications. It can produce strong vibrations in the 20 Hz to 80 Hz range and comes at a price range of around 19€.
- **Dayton Audio HDN-8 Weatherproof:** This is a high-performance, weatherproof tactile transducer designed for outdoor use. It can produce strong vibrations in the 20 Hz to 80 Hz range and comes at a price range of around 25€.
- **Dayton Audio DAEX25FHE-4:** This is a compact and affordable exciter with a frequency response of 70 Hz to 20 kHz and comes at a price range of around 15€.

For the sake of this project, the **Dayton Audio DAEX25FHE-4 exciter** was selected as the most suitable option for our stethoscope calibration system. This choice was based on its ability to produce strong vibrations in the relevant frequency range for cardiopulmonary sounds, as well as its affordability and compact size, which are important considerations for a practical calibration system.

Power Amplifier

To drive the Dayton Audio DAEX25FHE-4 exciter, the reference signal generated by the central control software must be passed through a power amplifier. The primary requirement for this component is signal linearity; the amplifier must deliver a perfectly flat frequency response from 20 Hz to 2000 Hz without introducing Total Harmonic Distortion (THD) or low-frequency roll-off.

Two primary amplifier topologies were evaluated for the physical test rig:

- **Class-D Amplifiers (e.g., PAM8403 / TPA3116):** Class-D modules operate utilizing PWM. While extremely affordable and capable of operating on

standard 5V USB power, the digital switching nature of Class-D topologies introduces high-frequency switching noise and poorer inherent linearity [7].

- **Class-AB Amplifiers (e.g., TDA2030 / LM1875):** Class-AB amplifier provides a continuous, highly linear analog output, ensuring a pure analog signal path across the entire target frequency spectrum. Literature demonstrates that Class-AB topologies consistently outperform Class-D stages in Total Harmonic Distortion (THD) metrics, frequently achieving distortion figures up to 30 dB better than their digital counterparts [8].

To ensure the acoustic reference signal remained free of high-frequency digital artifacts and low-frequency attenuation, the **Class-AB TDA2030** IC was selected. It comes at a highly cost-effective price point of around 2€.

The TDA2030 was then integrated into a power amplifier circuit to provide sufficient current to drive the Dayton Audio DAEX25FHE-4 exciter without introducing clipping or harmonic distortion.

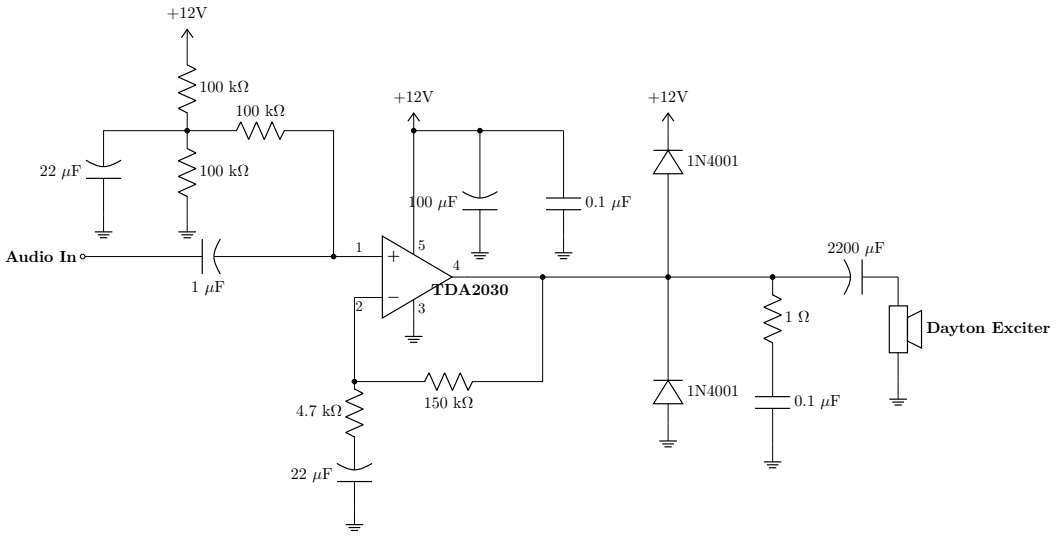


Figure 3.2: TDA2030 Module Schematic.

3.3.2 Sound Acquisition Technologies

To accurately capture the acoustic output of the stethoscope, the microphone must satisfy two critical requirements: it must possess a physical form factor small enough to achieve an airtight seal with the stethoscope's eartip (approximately 5 mm in diameter), and it must feature a flat frequency response capable of registering extreme low-frequency cardiovascular sounds down to 20 Hz.

Two primary microphone technologies were evaluated for the acquisition stage:

- **MEMS (Micro-Electro-Mechanical Systems) Microphones:** Commonly utilized in modern mobile devices, MEMS microphones are exceptionally compact and digitally integrated. However, consumer-grade MEMS capsules are specifically engineered for human speech transmission. Consequently, they frequently incorporate built-in high-pass filters designed to attenuate wind and handling noise below 60 Hz. Utilizing a standard MEMS microphone would result in the severe attenuation of foundational S1, S2, and S3 heart sounds, rendering them unsuitable for this calibration system.
- **ECM (Electret Condenser Microphone) Capsules:** ECMs are traditional analog acoustic sensors utilized in audio measurement. A standard 6 mm ECM capsule provides the necessary compact form factor for an airtight eartip coupling. More importantly, unlike consumer MEMS devices, measurement-grade ECM capsules do not artificially filter low-frequency kinetic energy, providing a highly linear frequency response down to 20 Hz.

Given the low-frequency requirements of cardiopulmonary auscultation, a 6 mm ECM capsule was selected as the optimal sensor for the acquisition stage. Because the raw analog output of an ECM capsule is highly attenuated (typically in the millivolt range), this selection necessitates the integration of a low-noise pre-amplification stage prior to digital signal processing.

Table 3.3: Comparison of microphone sensitivity and voltage output for different ECM capsules.

Microphone Example	**Sensitivity (dBV/Pa)**	**Voltage Output (mV/Pa)**
Generic Low-Cost ECM	-60 dB	1.00 mV
High-Sensitivity ECM	**-40 dB**	**10.00 mV**
Premium Measurement Mic	-20 dB	100.00 mV

As demonstrated in Table 3.3, microphone sensitivity operates on a logarithmic scale. A Generic Electret Condenser Microphone rated at -60 dB outputs approximately 1.00 millivolts per Pascal of acoustic pressure. By acquiring a higher sensitivity capsule rated at -40 dB, the electrical output increases drastically to 10 millivolts per Pascal, yielding a signal almost 10 times stronger for the exact same acoustic event.

This stronger signal reduces the gain requirements for the subsequent ADC preamplification stage, therefore minimizing the amplification of inherent electronic noise and improving the overall Signal-to-Noise Ratio (SNR) of the acquired data.

Pre-Amplifier

Despite selecting a high-sensitivity Electret Condenser Microphone (-40 dBV/Pa), the raw analog output generated by cardiopulmonary acoustics remains in the low millivolt range. To prevent quantization error and signal loss during analog-to-digital conversion, the microphone's output must be conditioned and boosted using a dedicated pre-amplifier stage.

To address the extremely low amplitude of physiological sounds, the transducer was interfaced with a MAX9814 microphone amplifier module. The MAX9814 was selected for its integrated low-noise preamplifier and Automatic Gain Control (AGC) circuitry. The AGC dynamically compresses loud ambient noises while maximizing sensitivity to faint biological signals, which is critical for unpredictable clinical environments. The module was powered via the 3.3 V rail of the microcontroller, and its gain pin was tied high to lock the amplification at 40 dB, providing an optimal balance between signal enhancement and ambient noise floor rejection.

Because cardiopulmonary acoustics are alternating current (AC) waveforms, the signal required a precise direct current (DC) bias to be accurately read by the single-supply analog-to-digital converter. The MAX9814 outputs a resting DC bias of 1.25 V. To safely center the waveform for the microcontroller, a voltage divider network was constructed using two 100 kΩ resistors between the 3.3 V supply and ground, establishing a stable 1.65 V to 1.8 V reference node.

A 10 μF electrolytic coupling capacitor was placed in series between the MAX9814 output and the voltage divider. This capacitor acts as a DC block, preventing the 1.25 V bias of the amplifier from colliding with the 1.8 V bias of the microcontroller node, while allowing the AC biological audio frequencies to pass through unimpeded.

Hardware Anti-Aliasing Filter

To prevent high-frequency environmental noise and digital aliasing from corrupting the target cardiopulmonary data, a passive resistor-capacitor (RC) low-pass filter was engineered directly at the microcontroller input. The target acoustic bandwidth for heart and lung sounds spans from 20 Hz to 2000 Hz. A 5.6 kΩ series resistor and a 10 nF ceramic capacitor were selected to construct the filter. The cutoff frequency (f_c) was calculated using the standard RC filter equation:

$$f_c = \frac{1}{2\pi RC} \tag{3.1}$$

$$f_c = \frac{1}{2\pi(5600\ \Omega)(10 \times 10^{-9}\ \text{F})} \approx 2842\ \text{Hz} \tag{3.2}$$

This 2842 Hz cutoff frequency acts as an acoustic window. It allows 100% of the 2000 Hz medical bandwidth to pass into the ADC mathematically untouched, while aggressively attenuating high-frequency room static and serial transmission noise.

3.3.3 Signal Digitization and Transmission

Following the analog signal conditioning stage, the acoustic data must be digitized for computational analysis. Signal digitization was executed using an **ESP32 microcontroller**, leveraging its internal 12-bit Analog-to-Digital Converter (ADC). This ADC resolution allows the biological waveform to be mapped across 4096 discrete voltage steps.

A critical limitation in digital auscultation is the Nyquist sampling theorem, which dictates that the sampling rate must be at least twice the highest frequency of interest to prevent data loss. To accurately visualize a 2000 Hz cardiopulmonary spectrogram, a minimum sampling rate of 4000 Hz was strictly required. Standard serial communication speeds create a data bottleneck, physically capping the sampling rate and blinding the software to the upper diagnostic frequencies. To overcome this limitation, the ESP32 serial baud rate was set to 500,000 bps, and the loop delay to 50 microseconds. This optimization successfully pushed the hardware sampling rate well above the 4000 Hz threshold, fulfilling the Nyquist criteria for the entire lung sound bandwidth.

The raw serial data was subsequently logged locally using a custom Python script utilizing the PySerial library, preparing the time-series data for spectral analysis.

Chapter 4

Results and Discussion

4.1 System Characterization via Controlled Frequency Sweeps

To evaluate the spectral response and operational limits of the integrated hardware prior to mechanical stethoscope coupling, a series of controlled sinusoidal tone sweeps were administered. The system's digital acquisition pipeline logged time-domain amplitude variations and converted the signals via Fast Fourier Transform (FFT) into dB-scaled spectrograms.

4.1.1 Low-Frequency Biomedical Bandwidth Analysis (20 - 50 Hz)

Biomarkers such as S1 and S2 heart sounds, alongside pathological S3 and S4 gallops, rely heavily on extreme low-frequency preservation. Testing the system at a fundamental frequency of 20 Hz verified the hardware's capacity to process foundational kinetic energy without artificial signal attenuation or low-frequency roll-off.

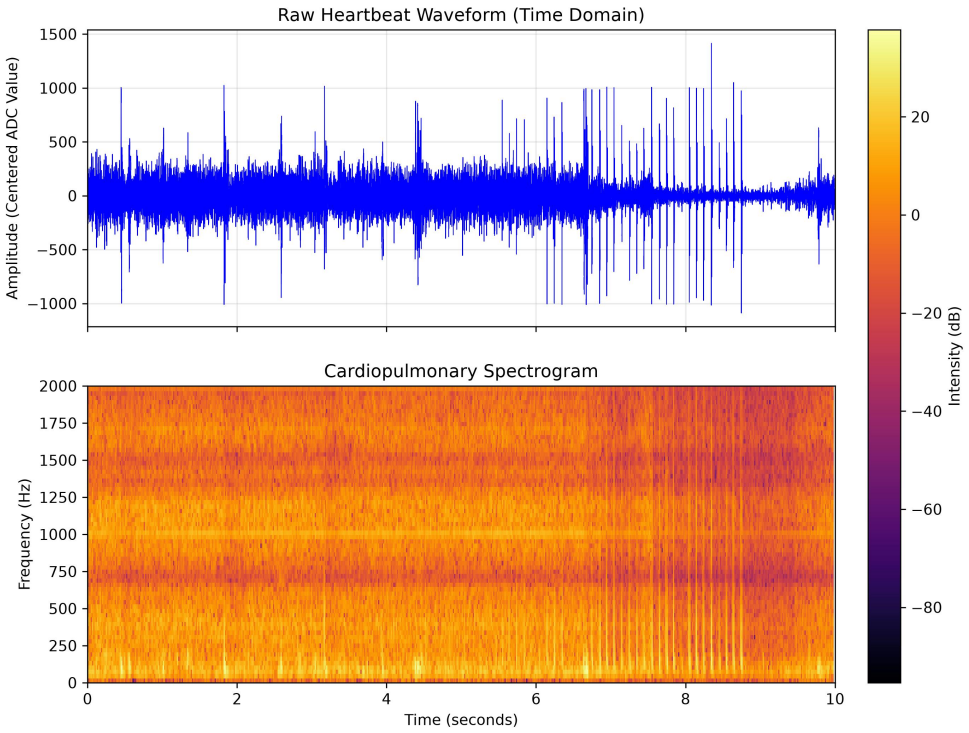


Figure 4.1: Time-domain waveform and corresponding spectrogram of a 20 Hz sinusoidal tone sweep. The waveform demonstrates clean periodicity with no DC offset, while the spectrogram confirms high-intensity energy capture at the 20 Hz frequency.

As demonstrated in Figure 4.1, the time-domain waveform while noisy, maintains a consistent amplitude envelope, and the spectrogram confirms that the system accurately captures the 20 Hz frequency with minimal harmonic distortion. However the sudden accuracy increase after 7 seconds is indicative of a noise floor reduction, likely due to the microphone's AGC (Automatic Gain Control) algorithm engaging after a period of low-amplitude input.

Extending the low-frequency evaluation to 50 Hz provides further insight into the system's tracking capabilities across the primary cardiovascular band.

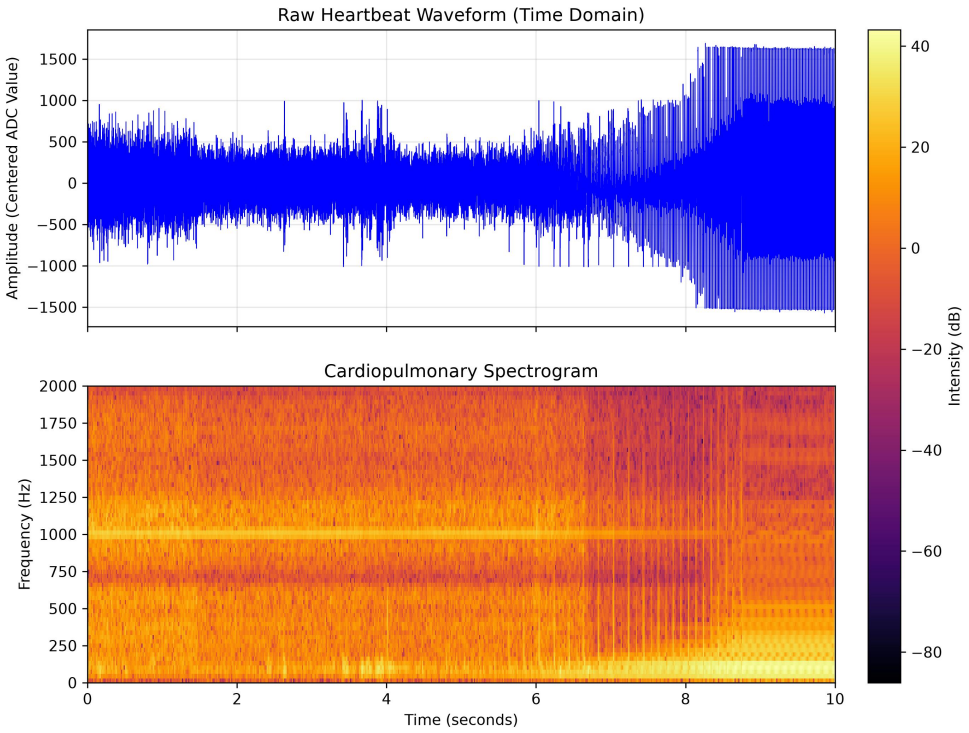


Figure 4.2: Time-domain waveform and corresponding spectrogram of a 50 Hz sinusoidal tone sweep. The waveform exhibits a stable amplitude envelope, while the spectrogram confirms a concentrated energy band at 50 Hz

The Figure 4.2 displays a dense, high-intensity band concentrated at the 50 Hz baseline. The time-domain envelope remains stable for the majority of the capture window before exhibiting a high-amplitude surge during the final two seconds. This surge pushes the waveform toward the 1500 ADC value threshold, resulting in a broader vertical dispersion of energy across the upper frequency tiers of the spectrogram.

4.1.2 Intermediate-Frequency Analysis (200 Hz)

Transitioning into intermediate frequencies, testing the system at 200 Hz evaluates the operational response and linearity of the amplification stage as the signal approaches the mid-range of the cardiopulmonary spectrum.

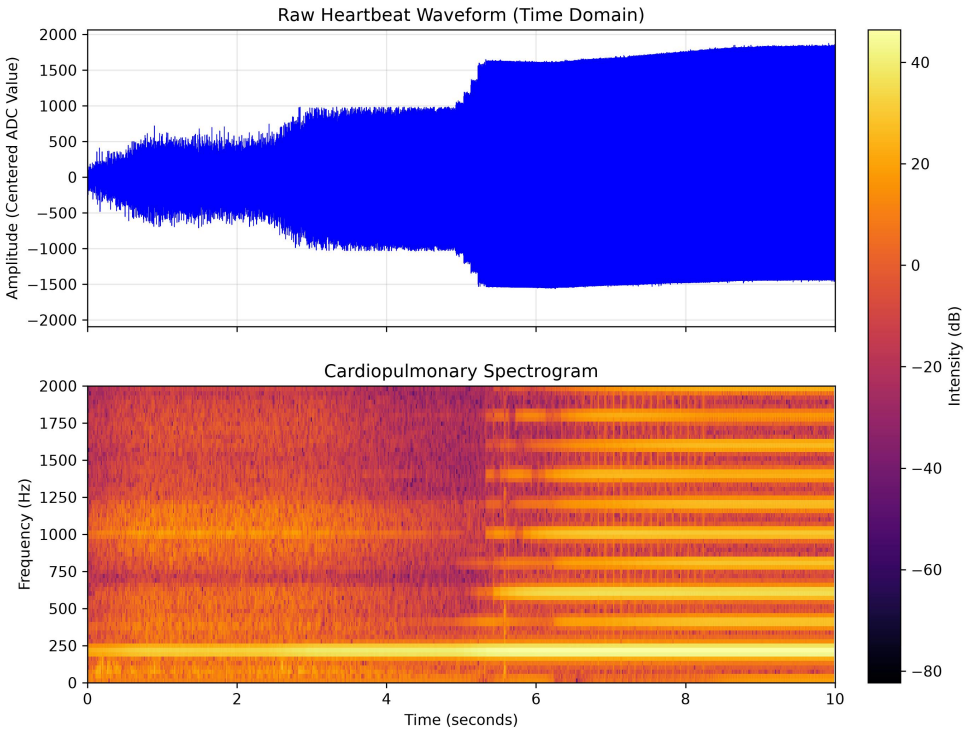


Figure 4.3: Time-domain waveform and corresponding spectrogram of a 200 Hz sinusoidal tone sweep. The waveform demonstrates a stable amplitude envelope with a sudden high-amplitude surge, while the spectrogram reveals harmonic distortion bands at integer multiples of the fundamental frequency.

As illustrated in Figure 4.3, the time-domain envelope remains relatively constrained during the first five seconds before experiencing a sharp, high-amplitude jump, expanding the waveform range toward ±1500 ADC units. Concurrently, the spectrogram exhibits a dense fundamental frequency band at 200 Hz accompanied by sharp, parallel horizontal intensity tracks stacking at exact integer multiples (400 Hz, 600 Hz, 800 Hz, 1000 Hz, and above). Rather than random system noise, these structured harmonic bands are indicative of Total Harmonic Distortion (THD).

4.1.3 High-Frequency Analysis (1111 Hz)

To verify that the digitization pipeline avoids the data bottlenecks typical of lower communication speeds and satisfies the Nyquist sampling theorem for the upper bounds of the cardiopulmonary bandwidth, high-frequency validation was performed using an 1111 Hz input tone.

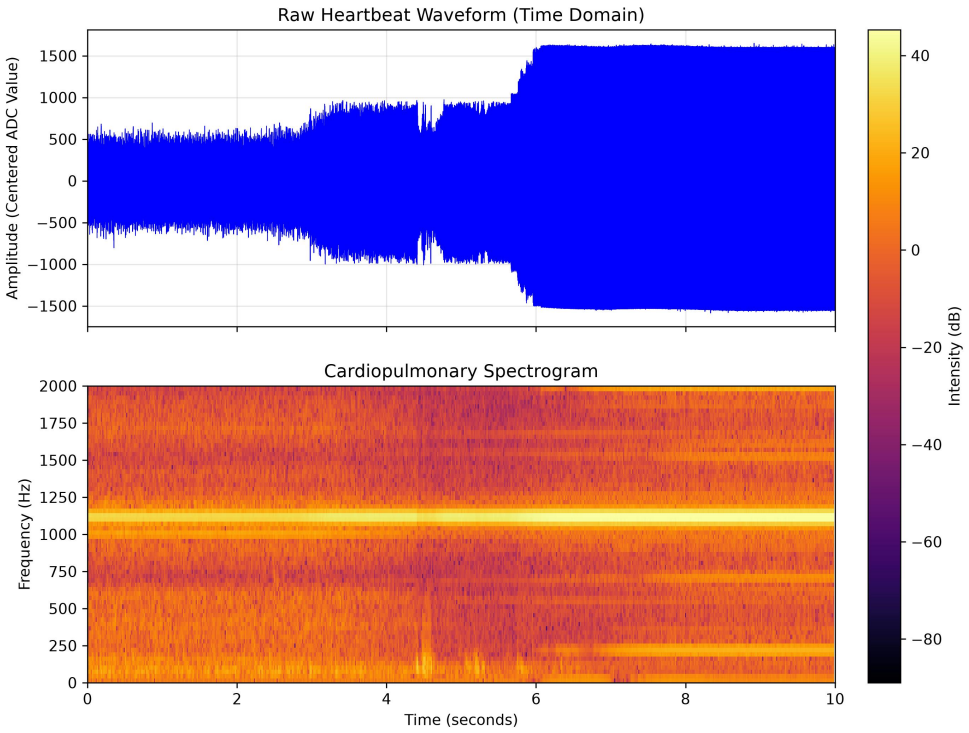


Figure 4.4: Time-domain waveform and corresponding spectrogram of an 1111 Hz sinusoidal tone test. The waveform demonstrates stable high-frequency acquisition, while the spectrogram confirms a continuous, clean energy band at 1111 Hz without attenuation or digital blindness.

The spectrogram in Figure 4.4 displays a sharp, continuous horizontal intensity band locked precisely at the 1111 Hz mark across the entire ten-second capture window. This visualization confirms that increasing the ESP32 serial communication baud rate to 500,000 bps successfully maintains a hardware sampling rate well above the 4000 Hz threshold required to map the 2000 Hz medical ceiling.

4.2 Conclusion and Future Work

This project designed, implemented, and validated an acoustic calibration and visualization system for stethoscopes. By integrating an ESP32 data acquisition pipeline, a MAX9814 pre-amplifier stage, a TDA2030 amplification module. The prototype captured and mapped the entire 20 Hz to 2000 Hz cardiopulmonary frequency spectrum. Discrete frequency testing confirmed the system's ability to isolate

low-frequency physiological events while maintaining high spectral fidelity. Ultimately, the developed architecture offers a scalable framework to evaluate stethoscope degradation and ensure clinical reliability. Currently the system is limited to single-channel acquisition, and future work will focus on implementing multi-channel support to enable simultaneous bilateral auscultation. Additionally, further research will explore the integration of machine learning algorithms for automated anomaly detection and classification of heart and lung sounds, enhancing the diagnostic capabilities of the system.

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